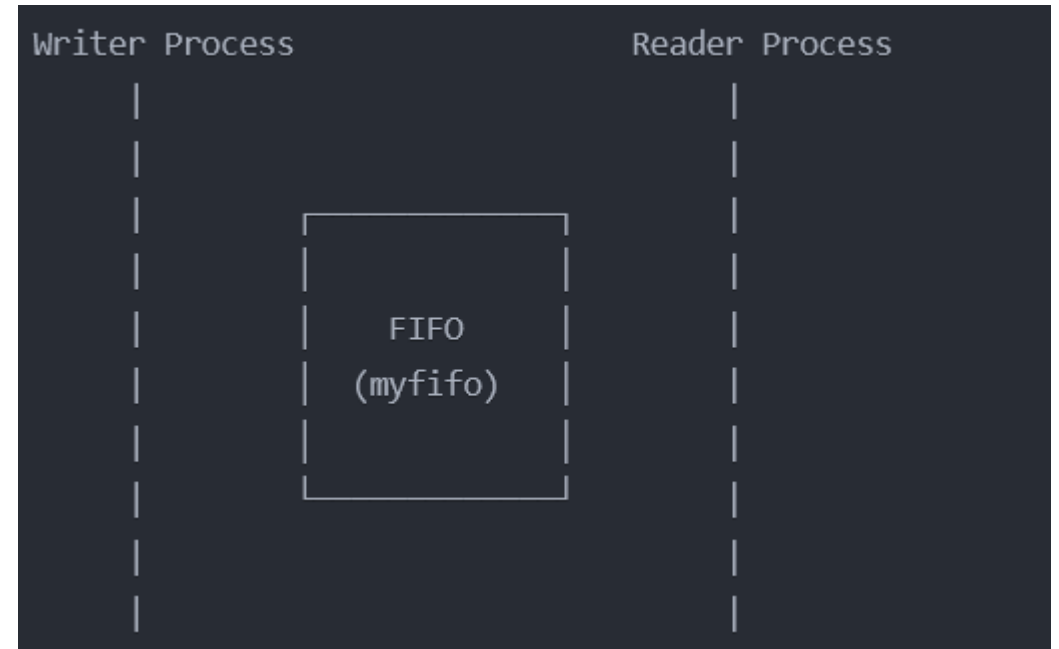


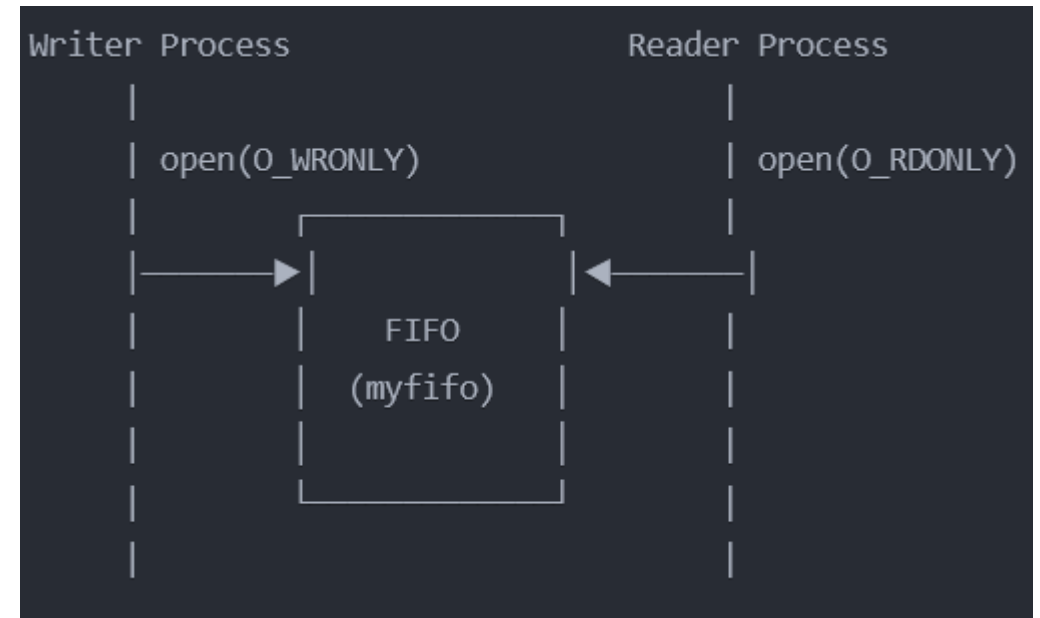
Initial Setup

- Both processes can see the named pipe in the file system.
- The named pipe exists independently of both processes.

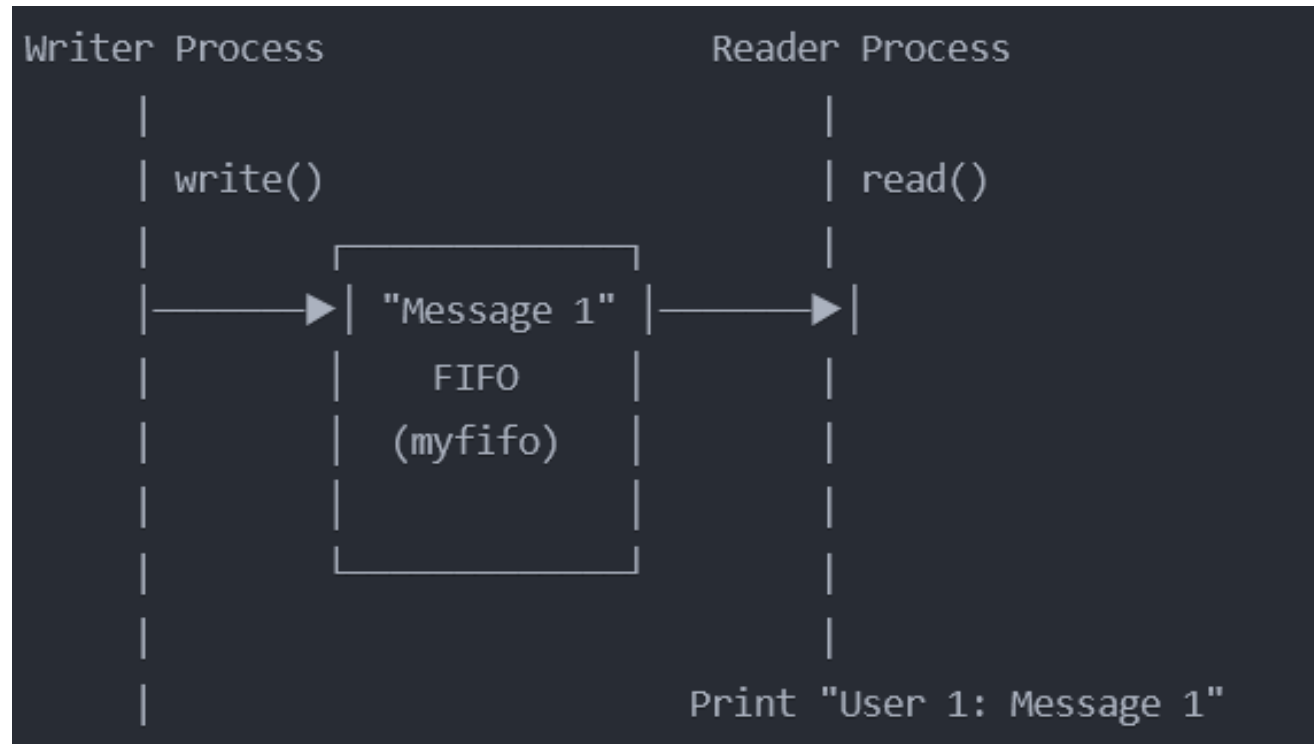


Communication Cycle

- Processes alternate between writing and reading.
- Processes open the named pipe in different modes (O_WRONLY or O_RDONLY) depending on whether they're writing or reading.
- One process must open the named pipe for writing before the other can open it for reading.

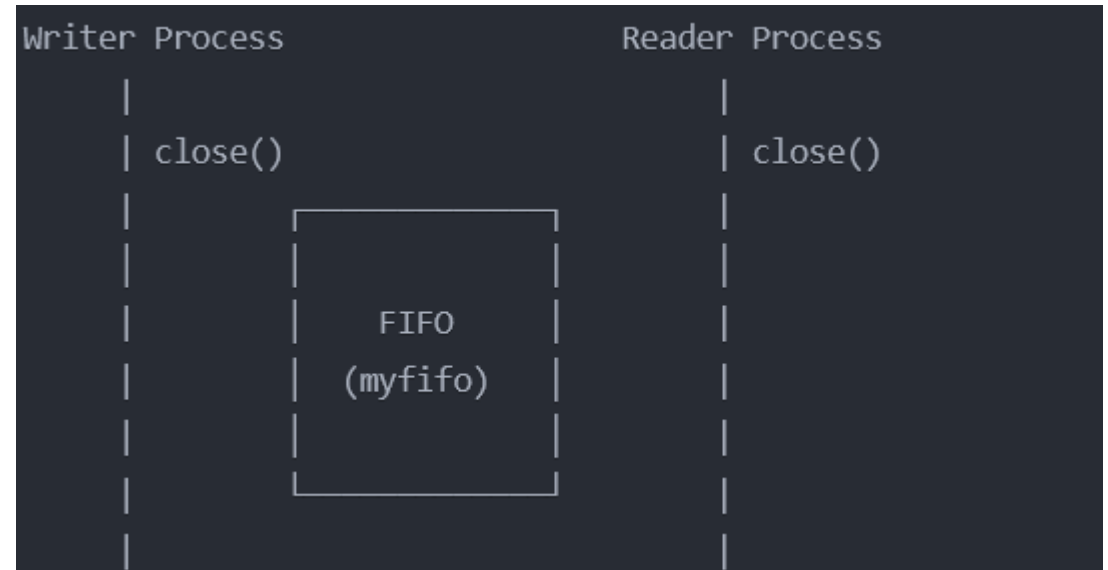


Writer writes, Reader reads

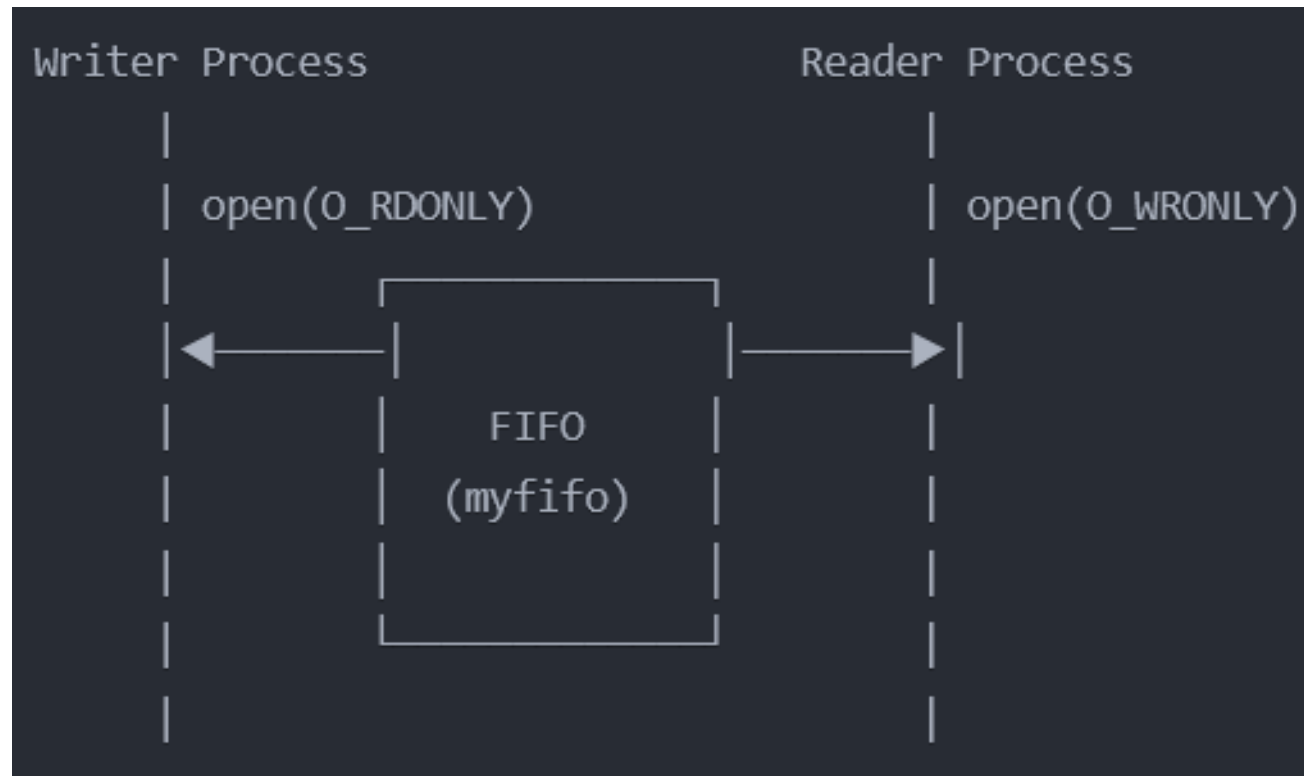


Writer closes write end, Reader closes read end

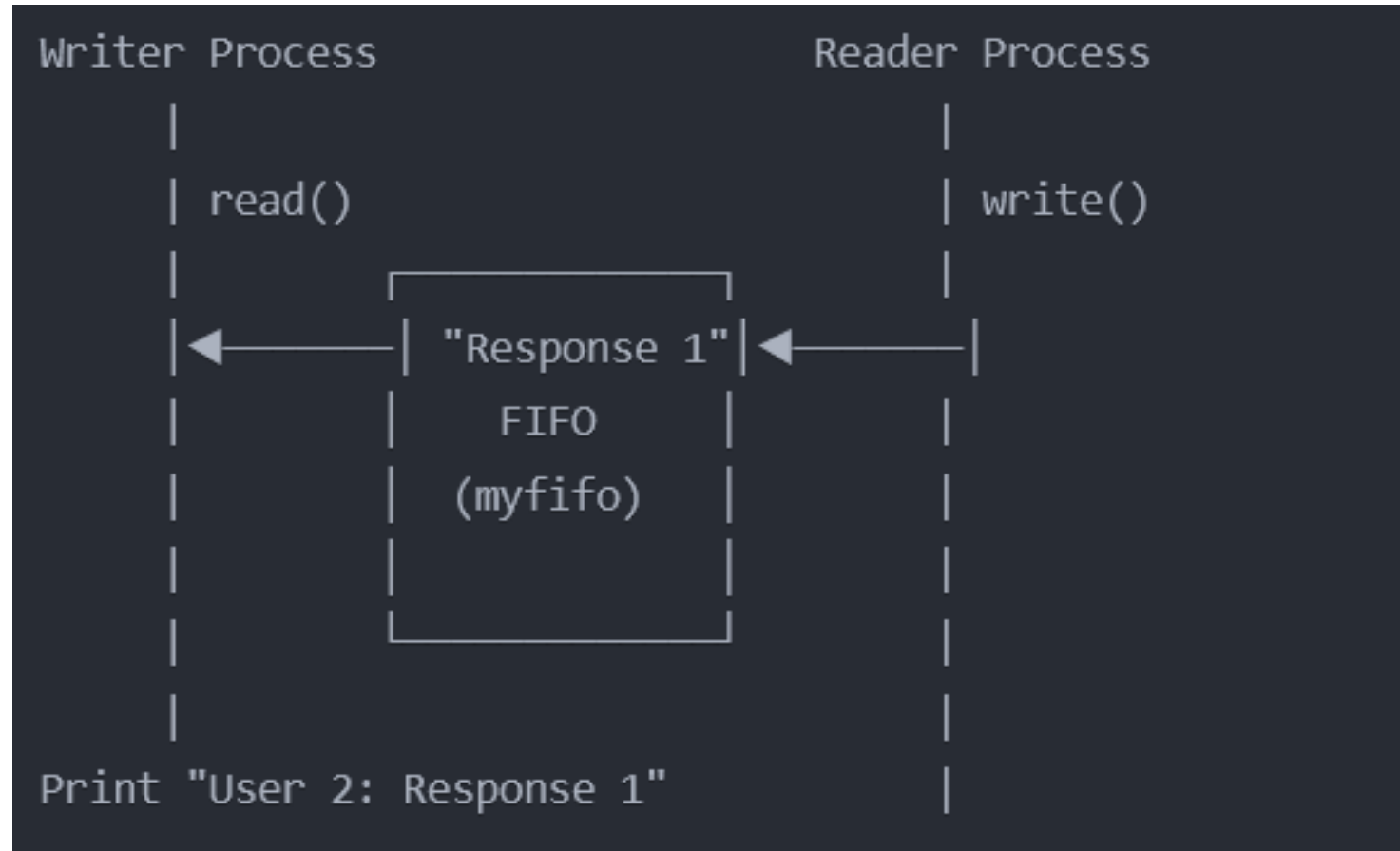
- Processes close their end of the FIFO after each operation



Reader opens FIFO for writing, Writer opens for reading



Reader writes, Writer reads



Writer closes read end, Reader closes write end

This cycle repeats, allowing for ongoing two-way communication.



Key Points

- Persistence: Unlike unnamed pipes, the FIFO persists in the file system and can be accessed by name.
- Synchronization: The open calls block until both ends of the pipe are opened, ensuring synchronization between processes.
- Alternating Access: The processes take turns writing and reading, facilitating a chat-like interaction.